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Negative News and Investor Trust: The Role of $Firm and #CEO Twitter Use

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ABSTRACT

We examine how CEOs can facilitate the development of investor trust that helps mitigate the effects of negative information. Results from an experiment show that investors trust the CEO more and are more willing to invest in the firm when the CEO communicates firm news followed by a negative earnings surprise through a personal Twitter account than when the news and surprise comes from the CEO via a website or from the firm's Investor Relations Twitter account or website. A follow-up experiment shows that repeating the negative news does not incrementally affect investors who received the news from the CEO's Twitter account, but does further negatively impact investors

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who received the news via other disclosure mediums, especially those who received the news via the Investor Relations Twitter account. Our results have implications for firms and executives considering the costs and benefits of communicating with investors via Twitter.

JEL codes: G11; G40; M41

Keywords: social media; Twitter; social bond; trust; investment decisions; negative earnings surprise; repeated information

1. Introduction

Firms increasingly use social media to disclose financial information (Joyce [2013], Blankespoor, Miller, and White [2014]), and investors increasingly use social media to retrieve firm information (Alden [2013], Brunswick [2014]). One of the most popular social media disclosure platforms is Twitter. Using Twitter to disclose financial information introduces two unique features not present in traditional disclosure platforms. First, Twitter creates an opportunity for investors to develop a social bond with whomever is tweeting out the information. Second, Twitter increases the repetition of information through multiple tweets and retweets of selected content (Miller and Skinner [2015]). We examine how these features have the potential to facilitate the development of investor trust that helps mitigate the effects of negative information.

The specific information we focus on is negative earnings surprises. We do so because prior research suggests these events can have significant consequences, for example, a disproportionately large stock price decline (Skinner and Sloan [2002]), loss of investor trust (Graham, Harvey, and Rajgopal [2005]), and negative career outcomes for the managers (Mergenthaler, Rajgopal, and Srinivasan [2012]). As a result, managing negative earnings surprises is important to managers (Graham, Harvey, and Rajgopal [2005]). Our study is important because it sheds light on how managers can mitigate investors' loss of trust and negative reaction that follows a negative shock. In the first of two experiments, we examine if investors develop more enduring trust in a firm's CEO when that CEO communicates with them via Twitter. We further examine whether this trust helps temper investors' reaction to subsequent negative firm news. In the second experiment, we examine if the trust investors developed is robust to the repetition of negative news. In addition, our second experiment has the incremental benefit of providing a potential explanation for the finding that the market is more sensitive to negative earnings surprises when firms are active on Twitter (e.g., Curtis, Richardson, and Schmardebeck [2016], Jung et al. [2017]).

To examine our research questions, we draw on social identity theory to predict that investors develop social bonds when another individual discloses information via Twitter. The social bond is created by the perception that investors are personally interacting with the other individual. A social

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bond refers to an individual's subjective sense of interpersonal closeness or connectedness with another (Kadous, Leiby, and Peecher [2013]). Social bonds generally cause individuals to develop more enduring trust in the other individual (Shapiro, Sheppard, and Cheraskin [1992], Lewicki and Bunker [1996], Haslam and Ellemers [2005]). Importantly, the extent to which the trust remains following negative news depends on the strength of the social bond developed; the stronger the initial bond the more likely trust remains intact after negative news.

We focus on trust at an operational level because (1) theory suggests social bonds are reflected in measures of trust and (2) trust has been shown to matter in an investment context (see, e.g., Elliott, Hodge, and Sedor [2012]). Since we focus on trust and the formation of a social bond, we measure the two foundations of trust—affective and cognitive trust (Lewis and Weigert [1985]). Mercer [2005, p. 724] examines a related, but distinct, construct of management credibility, which she defines as "investors' beliefs about management's trustworthiness and competence in financial disclosure." Credibility captures the cognitive foundation of trust. We measure that foundation as well as the affective foundation of trust, which is theoretically important for the formation of social bonds.

We expect that the source of a message, and the medium used to communicate the message, interact to affect the strength of the social bond that forms, which in turn affects investors' trust. Specifically, we expect that investors will develop a stronger social bond (and more enduring trust) when a CEO communicates firm news via a personal Twitter account than when the firm's Investor Relations communicates news via a Twitter account or when news is communicated via a website. As a result of having formed enduring trust, investor reactions to subsequent negative news will be tempered relative to when the same news is received via other disclosure mediums. We follow up our primary analysis by examining the mediating role trust plays in influencing investor decisions in our setting.

We test our predictions by conducting two experiments. In our primary experiment, we isolate one of the two key features of Twitter: its ability to facilitate a social bond. Doing so increases the internal validity of our experiment, but at the same time, does not mimic how Twitter functions in the real world. We therefore conduct a follow-up experiment that more closely mimics Twitter in the real world, thereby increasing the generalizability of our findings. Both experiments use 2 × 2 between-subjects designs with the same independent variables: disclosure medium (website vs. Twitter) and message source (Investor Relations vs. CEO).[1]

[1] Our study examines a potential benefit of CEOs using personal Twitter accounts to communicate firm news, but this choice could also introduce costs that we do not examine in our study. As discussed in more detail in section 2, several legal and regulatory risks exist related to social media accounts, including risks of violating advertising, intellectual property, and SEC laws and regulations (e.g., Regulation FD).

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In our primary experiment, participants assume the role of prospective investors evaluating a potential investment. All participants first view pictures of both the Investor Relations logo and CEO photo, then receive information about the firm followed by a series of firm press releases. We manipulate disclosure medium by varying whether participants receive links to the press releases via a website or Twitter. We manipulate message source by varying whether the news comes from the firm's Investor Relations or CEO. Our manipulations occur for all press releases, starting with the first press release and continuing until the final press release. After viewing the initial series of press releases, participants provide preliminary investment judgments. Participants then view the final press release via the same disclosure medium (website or Twitter) announcing that the firm has missed its earnings target due to an internal one-time event. Participants then make final investment judgments, respond to a series of trust questions, and complete the postexperiment questionnaire.

Results from our primary experiment support our predictions, and reflect that disclosure medium and message source have an interactive effect on investors' willingness to invest. Specifically, investors are more willing to invest in a firm when the CEO communicates firm news followed by a negative earnings surprise through a Twitter account than when the news and surprise come from the CEO via a website or from the firm's Investor Relations Twitter account or website. We also provide evidence that when a CEO establishes a social bond by communicating with investors through a Twitter account, investors exhibit more enduring trust. Finally, consistent with our theory, we show that trust mediates the relationship between disclosure medium and message source and investment decisions.

In our follow-up experiment, we broaden our focus to examine two features of Twitter that naturally occur together in the real world: the repetition of information and development of social bonds. Jointly examining these features increases the generalizability of our primary results by addressing whether the social bond that developed in the CEO Twitter condition is robust to the repetition of previously disclosed negative news. In our follow-up experiment, we expect the relative strength of the social bond between investors and a message's source (CEO or Investor Relations) to determine how investors react to repeated negative news. When a relatively strong social bond has formed, we expect repeating previously disclosed negative news to have little effect on investor judgments. In contrast, when a relatively weak social bond has formed, we expect repeating previously disclosed negative news will negatively affect investors.

Participants in our follow-up experiment receive the same information, but participants in Twitter conditions receive additional tweets repeating quotes taken from previously disclosed information. Consistent with our expectations, repeating previously disclosed negative news does not negatively impact investor trust or investment decisions when the message source is the CEO's Twitter account, but does negatively impact trust and investment

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decisions when the message source is the firm's Investor Relations Twitter account. Investors in this latter condition exhibit a low level of trust that has been eroded by the repeated negative news tweets from the Investor Relations Twitter account. Further, when a CEO communicates via Twitter, thereby establishing a strong social bond, investors are more likely to believe the firm's statement that the negative news is a one-time event. We also rule out an alternative explanation that the message source affects perceptions of the credibility of the information. Specifically, we provide evidence that an Investor Relations account is not perceived as more credible than a personal CEO account.

Our study makes several contributions. We contribute to recent literature investigating the consequences of social media (e.g., Sprenger et al. [2013], Blankespoor, Miller, and White [2014], Chen et al. [2014], Chen, Hwang, and Liu [2017]). We provide evidence that investors who receive multiple messages containing the same negative news from a firm's Investor Relations Twitter account are less willing to invest than investors who receive the same information from the firm's website. This finding suggests the repetition of information, an inherent feature of social media platforms like Twitter, may be a reason recent studies find higher market sensitivity to earnings surprises (e.g., Curtis, Richardson, and Schmardebeck [2016], Jung et al. [2017]).

We also contribute to the literature examining factors that influence the market's reaction to earnings news. Prior research finds that the market reacts more strongly to firm disclosures when earnings are persistent (Kormendi and Lipe [1987], Easton and Zmijewski [1989]), firm risk is lower (Easton and Zmijewski [1989]), firm growth opportunities are higher (Collins and Kothari [1989]), and credibility is higher (Teoh and Wong [1993], Hutton, Miller, and Skinner [2003], Rogers and Stocken [2005], Gu and Li [2007]). We extend this research in several ways. In our setting, the CEO provides a qualitative "soft talk" disclosure (i.e., Hutton, Miller, and Skinner [2003]) that a negative earnings surprise is of low persistence. Investors are more likely to believe that statement, and thus react less negatively to the surprise when they have enduring trust in the CEO, which is more likely the case when the CEO communicates with investors via Twitter. We also provide evidence that this effect persists even when the negative surprise is repeated. In contrast, investors are more likely to interpret the negative surprise as more persistent, and thus react more negatively to the surprise, when trust is less enduring. Our results show this is the case when the news comes from a website or the Investor Relations Twitter account.

Our study also extends the disclosure medium literature (e.g., Elliott, Hodge, and Sedor [2012]). We examine Twitter, which captures a different construct from other disclosure mediums (such as video or text) in that it provides the opportunity to develop a social bond with the message source through the perceived ability to interact with the source. This important conceptual distinction between video and Twitter mediums leads

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to four primary contributions of our study to the disclosure medium literature.[2] First, a strong social bond affects the way that investors interpret repeated negative news. Specifically, a strong social bond leads investors to discount repeated negative news. Second, we are the first to examine how different information sources interact with how negative news is disclosed. Examining this interaction is important because most firms communicate financial news via an Investor Relations Twitter account and our theory and results suggest that firms can minimize investors' negative reaction by disclosing news via a CEO Twitter account. Third, prior research examines how CEOs can use video to ex post repair damaged trust caused by significant negative events, like restatements (e.g., Elliott, Hodge, and Sedor [2012]). We show how CEOs can use Twitter to ex ante build enduring trust (strengthened through social bonds) that persists following important negative events that are more common than restatements (e.g., a missed earnings target). Finally, our study helps educate CEOs by demonstrating that they could reap potential benefits from establishing a Twitter account and simply tweeting already scripted information from press releases. This is important to demonstrate as many CEOs are reluctant to communicate with stakeholders via Twitter due to the perceived time commitment and potential legal ramifications of unscripted tweets going viral (Kwoh and Korn [2012]).

2. *Background and Hypothesis Development*

In 2012, Netflix CEO Reed Hastings used his personal Facebook page to disclose that Netflix had record online viewing hours the previous month. The post sparked controversy over whether Hastings violated Regulation FD and Section 13(a) of the Securities Exchange Act of 1934 by disclosing material nonpublic information on social media without an accompanying press release or Form 8-K (SEC [2013b]). In response, the Securities and Exchange Commission (SEC) issued a Report of Investigation under Section 21(a), which announced that it would not pursue an enforcement action against Hastings. Instead, the report clarified how Reg FD and the SEC's 2008 interpretive release regarding the disclosure of material information on company websites apply to social media. Specifically, firms should apply the 2008 Guidance to social media, meaning firms can use social media as "recognized channels" to disclose material nonpublic information as long as investors are alerted in advance about which social media outlets will be used to disseminate information (SEC [2008, 2013a, b]). The SEC further acknowledged that, as long as investors are notified

[2] Elliott, Hodge, and Sedor [2012] hold constant the source of negative news (a firm's CEO) and manipulate whether the news is disclosed via video or a press release, and whether the CEO takes responsibility for the negative event or blames others. The video medium examined in that study impacts investors because it is a richer communication venue than a text-based press release.

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in advance, personal social media sites of individual employees (like CEOs) can also be used as recognized channels. The SEC suggested that firms can alert investors by including a disclosure in SEC filings or on corporate websites identifying the specific social media channels the firm intends to use for the dissemination of material nonpublic information (SEC [2013b]).

Since that announcement, firms have increased their use of social media outlets to communicate with stakeholders (Barnes and Lescault [2011, 2014], SEC [2013a]). Some firms now follow the SEC's suggestion and provide a "social media disclosure" on their website and in SEC filings notifying investors that material nonpublic information might be disclosed via a social media account. They also provide links to the specific social media accounts that are used, thus establishing those accounts as recognized channels. Some firms, like T-Mobile and Netflix, link to the CEO's personal social media site (Netflix [2017], T-Mobile [2017]). Our study mimics an alternative Reg FD compliant practice currently followed by the majority of firms. Specifically, most firms use social media as a complementary medium by tweeting or posting a link to press releases or SEC filings followed by additional tweets or posts highlighting summary information (Blankespoor, Miller, and White [2014]). In this case, the original press release or SEC filing is considered the public disclosure of material information and any subsequent selective disclosure of that information—such as by a nonrecognized social media account—does not trigger Reg FD because that information, even if material, is already considered public (SEC [2008]).

We focus on the microblogging site Twitter because it is the number one social media tool for Investor Relations work (Joyce [2013], NIRI [2013]). Twitter users choose a profile picture and create a username preceded by the @ symbol (e.g., @TwitterIR), which becomes their Twitter identity. Users can send 280-character text messages or "tweets" (the limit was 140-character messages until September 2017), which are displayed on their profile, the home page of their followers, and in the Twitter public timeline. Every time a user Tweets, his or her username and profile photo are displayed next to the tweet.

Retail and institutional investors are also utilizing social media for investment activities (Brunswick [2015], Greenwich Associates [2015]). Investors can sign up for Twitter and follow firms they are either considering as an investment or have already invested in. Investors can also view tweets from firms even if they do not have their own Twitter account by going directly to Twitter's website or through other outlets that relay Twitter information.[3] For example, Bloomberg incorporates tweets into its data service. Traders and other investment professionals can monitor social media discussions about a firm directly on their Bloomberg terminal (Alden [2013]). Google

[3] Investors may also be inadvertently exposed to firm information via Twitter as 86% of social media users use Twitter to access and read general news, which often contains information about specific firms (Rosenstiel et al. [2015]).

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also incorporates tweets into search results as soon as the tweets are posted (Frier [2015]) and investors can view tweets in some brokerage accounts (Businesswire [2015]). Finally, archival research suggests social media use affects both market returns (e.g., Sprenger et al. [2013], Chen et al. [2014], Chen, Hwang, and Liu [2017]) and liquidity (e.g., Blankespoor, Miller, and White [2014]).

Despite the growth in social media use by many market participants, many CEOs do not directly communicate with stakeholders via social media. In 2016, only 40% of Fortune 500 CEOs had a profile on at least one social media site, and only 7% had an active Twitter profile, up from 2% in 2012 (CEO.com [2012, 2016]). This low adoption rate could reflect the potential legal and regulatory costs CEOs face when they have personal social media accounts and use them to discuss their firms' results. Surveys of executives support this conjecture, and indicate that the primary reasons executives are hesitant to adopt social media are that participating is risky and takes too much time, social media use is not typical for their region or industry, and the belief that participating has no measurable return on investment (Kwoh and Korn [2012], Weber Shandwick [2014]).

Some risks of social media use are illustrated in firms' social media policies. Such policies have been adopted by many firms to guide online communication for employees (e.g., 77% of companies surveyed in 2016 by Barnes and Daubitz [2017] had a social media policy). Some companies (e.g., General Motors) have policies prohibiting communicating on behalf of the company or creating new social media accounts without approval. Other companies (e.g., Dell) have policies encouraging employees to use social media to champion the company. While social media policies differ across firms, many touch on the same legal and regulatory risks, risks associated with violating advertising, intellectual property, and SEC laws and regulations (Boudreaux [2015], Culp, Gomes, and Narveson [2015]). For example, Intel's policy highlights the risk of violating the Federal Trade Commission's laws, which could occur when comparing products to competitors' products without the appropriate disclaimers. Coca-Cola's policy highlights the risk of violating intellectual property laws, which could occur when sharing or retweeting content without giving credit to the content's creator. Finally, Best Buy's policy outlines the risk of violating SEC regulations like Reg FD, which could occur when nonrecognized users disclose nonpublic financial or operational information. The 2012 Netflix CEO Reed Hastings case illustrates the concern firms have about violating SEC regulations. Hastings's personal social media account was not considered a recognized disclosure channel for Netflix, so the SEC took notice when his tweets discussing material information were not accompanied by the original press release or SEC filing as required by Reg FD.

Accounting and consulting firms frequently provide guidance for managing social media risks that exist for both firms and CEOs, including legal and regulatory compliance risks, security risks, and reputational risks. However, overwhelmingly these groups claim that the most significant

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reputational risk related to social media is abstaining from social media. These groups recommend mitigating risks by monitoring the conversations stakeholders are having about them on social media and responding quickly when issues emerge (Ernst and Young [2014], Deloitte Touche Tohmatsu [2015], KPMG [2015], PwC [2017]). Recent research supports that firms can mitigate reputational damage by engaging in, rather than abstaining from, conversations on Twitter (Lee, Hutton, and Shu [2015], Cade [2017]).

While the issues above relate to both firms and CEOs, a recent survey of Fortune 500 employees highlights a potential benefit of CEOs engaging with social media. Eighty-two percent of survey respondents stated they are more likely to trust a company when the CEO engages with social media (Brandfog [2012]). Consistent with this survey evidence, psychology research suggests CEOs who disclose information via personal social media accounts likely create social bonds with message recipients, which facilitates the development of enduring trust.

2.1 THE INFLUENCE OF DISCLOSURE MEDIUM ON SOCIAL BONDS

We draw on social identity theory to examine how investors are influenced by social bonds created through disclosure mediums like Twitter. A social bond refers to an individual's subjective sense of interpersonal closeness or connectedness with another (Kadous, Leiby, and Peecher [2013]). Theoretically, several factors can create social bonds between individuals, including interpersonal interaction (Ashforth and Mael [1989]). Consistent with theory, prior research in accounting has manipulated social bonds by allowing participants to interact in experiments (e.g., King [2002], Kachelmeier and Van Landuyt [2017]).

It is important to note that *actual* interactions are theoretically not necessary to create social bonds; social bonds are a perceptual construct. An individual need only perceive that interaction is possible to form a social bond (Ashforth and Mael [1989]). We extend prior research by documenting that social bonds can develop simply from *perceptions* of interaction, which is a unique feature of disclosure mediums like Twitter (Short, Williams, and Christie [1976], Kaplan and Haenlein [2010]). Even if the option to interact is not exercised, the investor still likely forms an expectation that personal engagement with the CEO or firm is plausible (Lee and Shin [2014]). This interactive plausibility is not inherent through traditional communication mediums like a firm's website. As a result, investors are more likely to develop social bonds when using the Twitter medium relative to the website medium.

2.2 LEVELS OF SOCIAL BONDS CREATED BY MESSAGE SOURCE

Social identity theory further suggests that individuals tend to develop social bonds at various levels, ranging from low to high. Low-level social bonds tend to be stronger and are related to individuals and small groups, while high-level social bonds tend to be weaker and are related to whole

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organizations or divisions within organizations (Ashforth and Johnson [2001], for meta-analytic support, see Riketta and Van Dick [2005]). As a result, we expect that investors will develop a low-level, stronger social bond when the CEO communicates news via Twitter, and a high-level, weaker social bond when the firm's Investor Relations communicates news via Twitter. Finally, given a firm's website provides for limited or no interactions, we do not expect a social bond to form when the CEO or firm communicates via a website. As we discuss in the next section, social bonds and the enduring trust they facilitate likely influence investors' reactions to negative news.

2.3 REACTION TO A NEGATIVE EVENT: THE INTERACTION OF DISCLOSURE MEDIUM AND MESSAGE SOURCE

Trust is a psychological state in which one accepts vulnerability based on positive expectations regarding the intentions or behaviors of others (Mayer, Davis, and Schoorman [1995], Rousseau et al. [1998]). Social bonds help create a more enduring trust (Shapiro, Sheppard, and Cheraskin [1992], Lewicki and Bunker [1996], Haslam and Ellemers [2005]), which is likely to influence how investors react to negative news. Prior research in accounting has examined how managers can temper investors' reactions to significant negative events (e.g., a restatement), by ex post apologizing for the event and accepting responsibility by making an internal attribution. This research also shows that managers can repair trust to a greater extent when they apologize via a rich medium like video relative to text (Elliott, Hodge, and Sedor [2012]). We extend this research by examining how managers can temper investors' reactions to important negative events that are more common than restatements (e.g., missing an earnings target) by ex ante helping investors develop more enduring trust that persists following negative events (McAllister [1995], Lewicki and Bunker [1996], Morrison and Robinson [1997]).

We examine the impact of social bonds and the enduring trust they help create in a negative news setting for several reasons. First, negative events increase the importance of management's trustworthiness to investors and increase the extent to which trust influences investment decisions (Elliott, Hodge, and Sedor [2012]). Second, theory predicts that our independent variables, disclosure medium and message source, will interact to affect the *endurance* of trust, but not necessarily the *level* of trust. To elaborate, when a CEO or firm reports neutral or positive news, investors will develop similar levels of trust, but to varying degrees of endurance. The endurance of trust becomes important when the CEO or firm subsequently reports negative news. Investors who previously developed a social bond and more enduring trust by communicating with the CEO via Twitter, are more likely to maintain their trust in the CEO and believe an explanation he provides for the negative event (McAllister [1995], Jones and George [1998], Williams [2001]). Those who received information via a disclosure medium that does not facilitate the development of a social bond, and thus have less enduring trust, are more likely to question their trust and any

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explanation given for the negative event (Ashforth and Mael [1989], Ashforth, Harrison, and Corley [2008]). We test these expectations, and the important role trust plays in their development, in the following hypotheses:

H1: Investors will be more willing to invest in a firm when the firm's CEO communicates firm news followed by a negative earnings surprise through a Twitter account than when the news and surprise comes from the CEO via a website or from the firm's Investor Relations website or Twitter account.

H2: Trust will mediate the influence of disclosure medium and message source on investors' willingness to invest in the firm.

3. *Description of Primary Experiment*

3.1 PARTICIPANTS

Participants are 151 senior accounting students from a large university. Participants have taken on average 8.3 accounting courses and 2.3 finance courses. All but 10 participants have used financial statements to evaluate a firm's performance at least once. Thirty percent of participants have purchased common stock or debt securities, while 74% plan to do so in the next five years. We chose senior accounting students for several reasons. First, we match participants' knowledge to our task without using more sophisticated subjects than necessary (Libby, Bloomfield, and Nelson [2002]). Second, our participants have reasonable familiarity with social media. Finally, theoretically, age or work experience should not interact with the development of a social bond and, in turn, trust. Previous research has found students (e.g., Webber [2008]) and experienced managers (e.g., McAllister [1995]) develop trust in similar ways. We also confirm our primary results are inferentially identical using MTurk participants as an alternative proxy for nonprofessional investors. We discuss this replication in section 4.5. We provide a timeline of events for our experiment in figure 1.

3.2 DESIGN

Our experiment uses a 2 × 2 between-subjects design, with disclosure medium and message source as our manipulated independent variables. We manipulate disclosure medium by providing a link to firm press releases from the firm's website or from a Twitter account. We manipulate message source by providing the information from the firm's Investor Relations or CEO. We manipulate both disclosure medium and message source starting with the first press release in a series of press releases designed to allow the formation of a social bond and, in turn, trust. Participants view all press releases, including the final press release that reveals a negative earnings surprise, within their respective condition. The appendix includes excerpts from our experimental materials showing how we manipulated disclosure medium and message source.

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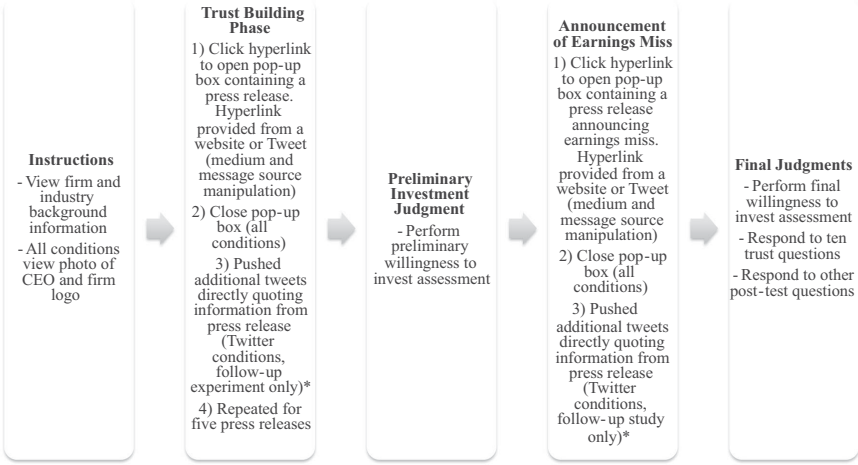


FIG. 1.—Timeline of events. This figure presents the timeline of events in the experiment. The only difference between the primary experiment and follow-up experiment is the asterisked steps. Participants in the primary experiment (Twitter conditions) are *not* pushed additional tweets. Instead, these participants move from Trust Building Phase step 2 to step 4, and move from Announcement of Earnings Miss step 2 to Final Judgments. Participants in the follow-up experiment are pushed additional tweets directly quoting information from the press release.

3.3 TASK AND PROCEDURES

Participants begin the experiment by accessing a hyperlink on their own computers. When they click on the link, they are randomly assigned to one of four experimental conditions. We instruct participants to assume they own a portfolio of stocks. As part of their annual review, they are considering investing in the confectionery industry to maintain a diversified portfolio. One of the firms they are considering is Armano.[4] We inform participants that they will view both financial and nonfinancial information about Armano as events occurred and as information was released, and that they will use this information to make their investment decision. Participants in all conditions begin the experiment by reviewing background information and images of Armano's logo and the firm's CEO, Dan Athens. Through the lens of the disclosure medium of their respective conditions, they then view a series of press releases designed to provide financial and nonfinancial information about Armano, its activities, and its management.

3.4 DISCLOSURE MEDIUM MANIPULATION

We manipulate disclosure medium by providing hyperlinks to all press releases from either the firm's website or Twitter. Participants in website

[4] We adapted experimental materials from Elliott, Hodge, and Sedor [2012]. The materials were constructed from press releases, Forms 8-K and 10-K, and Internet websites for companies operating in the confectionery and retail industries.

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conditions must click a link on their computers to view a pop-up box containing each press release, mimicking the process of accessing financial information via an Investor Relations website. Participants in Twitter conditions receive an initial tweet on their computers with a link to a pop-up box containing the first press release. Participants in all conditions can move to the next page only after they click the hyperlink to access each press release. This feature holds constant the effort required to access each press release and the information each participant views.

After participants close each press release pop-up box, they can move to the next page. In order to hold as much constant across conditions as possible, Twitter participants are not able to reply to any of the tweets. In the initial phase of our experiment, where participants have the opportunity to develop trust from a social bond, all but one of the press releases highlight positive information. The one that does not highlights that Armano intends to close underperforming stores. This press release builds trust through a social bond because Armano is being transparent about its intentions, the logic behind the closures, and its plans for helping all employees affected.

3.5 MESSAGE SOURCE MANIPULATION

We manipulate message source by making it salient that the links to the press releases are coming from either Armano's Investor Relations or the firm's CEO, Dan Athens. Website/Investor Relations condition participants view the press releases on a webpage with a picture of the firm logo salient and the phrase "Armano Investor Relations" next to the photo. Twitter/Investor Relations condition participants view the press releases through a Twitter account named "Armano Investor Relations" with Twitter handle @armano and a profile picture of the firm logo. Website/CEO condition participants view the press releases on a simple webpage with a picture of the CEO salient and the phrase "Armano CEO, Dan Athens" next to the photo. Twitter/CEO condition participants view the press releases through a Twitter account named "Armano CEO, Dan Athens" with Twitter handle @DanAthens and a profile picture of the CEO.

3.6 PRELIMINARY INVESTMENT ASSESSMENT

After participants view the press releases in the trust-building phase, they perform an initial willingness to invest assessment of Armano. We measure initial willingness to invest using two questions. First, we ask, "How attractive is an investment in Armano stock?" Participants respond on a 101-point scale with endpoints 0 ("very unattractive") and 100 ("very attractive"). Second, we ask, "How likely are you to invest in Armano stock?" Participants respond on a 101-point scale with endpoints 0 ("very unlikely") and 100 ("very likely").

3.7 ANNOUNCEMENT OF EARNINGS EXPECTATION MISS

Participants then view a final press release announcing earnings for the following year. Participants view this final press release within the same

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disclosure medium and message source condition they viewed the initial set of press releases. The final press release indicates "sales and earnings fell significantly short of our, as well as the market's, expectations . . . This was largely the result of customer service and order fulfillment problems . . . As we look to the new year, we expect to resolve the remaining order fulfillment issues and restore our service levels to our customers' satisfaction."

3.8 FINAL INVESTMENT ASSESSMENT

Participants make their final willingness to invest assessment by responding to the same two questions they answered when making their preliminary willingness to invest assessment. We remind participants their answers should reflect their final assessment and not how they would adjust their preliminary assessment.

3.9. TRUST MEASURES AND POSTEXPERIMENTAL QUESTIONS

After making their final investment assessments, participants respond to 10 questions designed to measure the two foundations of trust we expect to develop through a social bond—affective and cognitive trust (Lewis and Weigert [1985]). We focus on trust at an operational level because (1) theory suggests social bonds are reflected in measures of trust and (2) trust has been shown to matter in an investment context (see, e.g., Elliott, Hodge, and Sedor [2012]). Cognitive trust is a judgment based on evidence of another's competence and reliability (McAllister [1995], Webber [2008]). Affective trust is not based on outcomes, but is based on interpersonal bonds developed from frequent interactions (McAllister [1995]). We present our trust questions in table 2, panel A. We adapt our trust questions for an investment context from the validated scale in McAllister [1995]. Participants respond to all trust questions on a 101-point scale with endpoints 0 ("strongly disagree") and 100 ("strongly agree").[5] To conclude, participants answer two manipulation-check questions and respond to other control and demographic questions.

4. Results

4.1 MANIPULATION CHECKS

To assess the effectiveness of our disclosure medium manipulation, we ask, "How did the firm disclose its financial information?" with answer choices "the firm sent tweets via Twitter and provided links to the financial information" or "the firm only provided links to the financial information."

[5] We asked our trust questions only once at the end of our experiment to avoid demand effects (see Elliott, Hodge, and Sedor [2012] for a similar design). General propensity to trust, measured separately from our affective and cognitive trust questions, does not differ by condition.

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Eighty-nine percent of participants correctly answer this question. To assess the effectiveness of our message source manipulation, we ask, "Which picture was shown every time the firm issued a press release?" with answer choices "Armano CEO, Dan Athens" next to a photo of Dan Athens or "Armano Investor Relations" next to a photo of the firm logo. Eighty-four percent of participants correctly answer this question. There are no significant differences in correct response rates across conditions for either question (all p-values > 0.27, two-tailed).[6]

4.2 TEST OF H1

H1 predicts that investors will be more willing to invest in a firm when the firm's CEO communicates firm news followed by a negative earnings surprise through a Twitter account than when the news and surprise comes from the CEO via a website or from the firm's Investor Relations website or Twitter account. We measure willingness to invest by combining participants' final ratings of (1) the attractiveness of an investment in the firm's stock and (2) the likelihood that they would invest in the firm's stock. Higher values indicate higher willingness to invest. A Cronbach's alpha of 0.93 for our two final ratings confirms that our investment-related questions are capturing the same underlying construct.[7]

Table 1, panel A, reports cell sizes, means, and standard errors for participants' preliminary and final willingness to invest, and least square means and standard errors for our dependent variable, final willingness to invest adjusted for preliminary willingness to invest as a covariate. Figure 2, panel A, displays the pattern of cell means for our dependent variable. There is no significant difference in participants' preliminary willingness to invest across conditions.[8] Our experimental design does not follow a traditional pretest–posttest design because participants in all conditions respond to the preliminary willingness to invest question after viewing both of our experimental manipulations. We made this design choice because, theoretically, social bonds, and thus enduring trust, develop following multiple positive interactions with another. We ask preliminary willingness to invest questions after these positive interactions, and final willingness to invest questions after presenting participants with negative news. It is participants'

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[6] Our results are inferentially identical if we exclude participants who fail either manipulation check.

[7] Results are inferentially identical if we use either of the individual investment questions rather than the combined measure. Contrast test results are also robust to (1) using the change in willingness to invest as the dependent variable and (2) using final willingness to invest as the dependent variable *without* including preliminary willingness to invest as a covariate. Our two preliminary ratings have a Cronbach's alpha of 0.92.

[8] Shapiro–Wilk's tests indicate preliminary willingness to invest is strongly negatively skewed. To assess the robustness of our results, we apply a transformation of [square root (101 – willingness to invest)] for both preliminary and final willingness to invest. We report results of the untransformed values for H1 and H2 for ease of interpretation, but all results are inferentially identical if we use the transformed values.

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TABLE 1

Primary Experiment: How Disclosure Medium and Message Source Affect Willingness to Invest: Tests of H1

Panel A: Unadjusted and adjusted willingness to invest, mean (standard error), $n = 151$

	Experimental Condition			
	Website/IR ($n = 36$)	Website/CEO ($n = 51$)	Twitter/IR ($n = 30$)	Twitter/CEO ($n = 34$)
Unadjusted preliminary willingness to invest	67.92 (2.46)	71.86 (2.31)	71.83 (3.25)	74.41 (2.58)
Unadjusted final willingness to invest	48.61 (3.19)	50.20 (2.69)	49.50 (4.00)	58.38 (2.99)
Adjusted final willingness to invest	49.45 (3.20)	50.11 (3.28)	49.42 (3.48)	57.70 (3.28)

Panel B: ANCOVA model of final willingness to invest

Source of Variation	SS	df	MS	*F*-Statistic	*p*-Value
Disclosure medium	513.91	1	513.91	1.41	0.24
Message source	718.24	1	718.24	1.98	0.16
Disclosure medium × Message source	526.68	1	526.68	1.45	0.12†
Preliminary willingness to invest	2,078.15	1	2,078.15	5.72	0.02
Error	53,053.97	146	363.38		

Panel C: Planned contrast coding and follow-up simple effects tests for final willingness to invest

Source of Variation	df	*F*-Statistic	*p*-Value
Overall tests:			
Twitter/CEO greater than website/CEO, website/IR, or Twitter/IR. Contrast weights (3, –1, –1, –1)	1	4.59	0.02†
Follow-up simple effects tests:			
Effect of message source given Twitter disclosure medium	1	3.00	0.04†
Effect of message source given website disclosure medium	1	0.03	0.87
Effect of disclosure medium given firm Investor Relations message source	1	0.00	0.99
Effect of disclosure medium given CEO message source	1	3.22	0.04†

This table presents descriptive statistics, an ANCOVA model, and simple main effects tests for participants' willingness to invest. Participants rated (1) the attractiveness of an investment in the firm's stock and (2) the likelihood that they would invest in the firm's stock. The willingness to invest variable is simply an average of these measures. Our dependent variable, adjusted final willingness to invest, is final willingness to invest adjusted for preliminary willingness to invest as a covariate. The simple effects and contrast *F*-statistic are calculated using the ANCOVA error term and adjusted means. In our experiment, we manipulate disclosure medium by providing a link to firm press releases directly from a website or from Twitter. We manipulate message source by providing the information from Investor Relations (IR) or the CEO. †*p*-values are one-tailed equivalents for directional predictions.

reactions to the negative news that theory argues will differ across our conditions. This untraditional design choice biases against finding our predicted pattern of results. Specifically, support for H1 requires a difference in final willingness to invest after controlling for participants' preliminary willingness to invest even though both willingness to invest measures were

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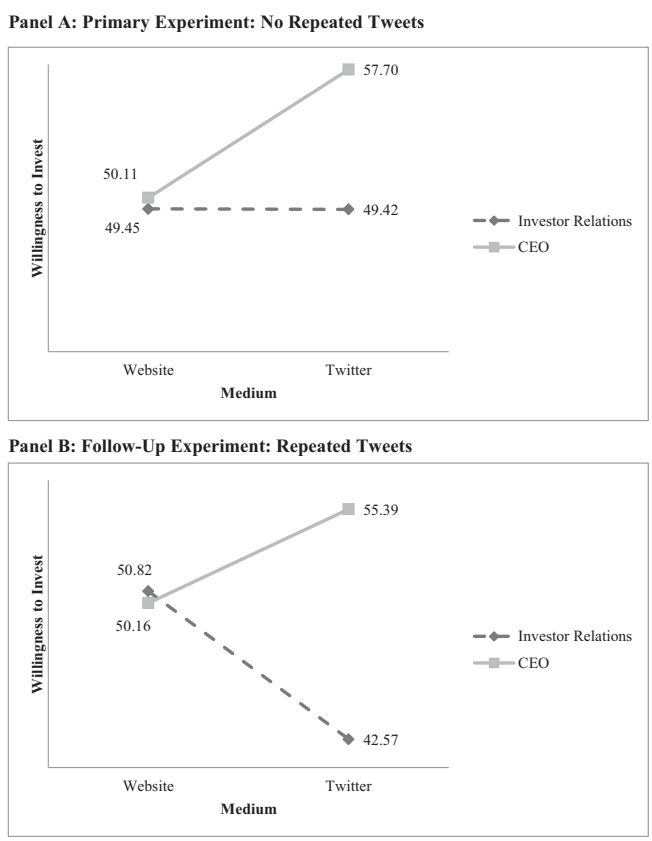


FIG. 2.—The effect of disclosure medium and message source on investors' willingness to invest. This figure graphically depicts our observed mean values for participants' final willingness to invest (adjusted for preliminary willingness to invest as a covariate). Panel A (panel B) depicts means for the Primary Experiment (the follow-up experiment) in which our Twitter manipulation does not (does) contain repeated tweets. Table 1 (table 4) presents descriptive statistics for preliminary and final willingness to invest for the Primary Experiment (the follow-up experiment).

captured after our manipulations. Thus, finding support for H1 provides strong evidence that participants reacted differently to the negative news depending on their experimental condition.

Table 1, panel B, presents ANCOVA results, and table 1, panel C, presents contrast test results and follow-up simple effects tests. H1 predicts an ordinal interaction, so using contrast codes according to our theoretically predicted pattern is the most appropriate way to test H1 as it improves statistical power without increasing Type 1 error rates (Buckless and Ravenscroft [1990]). Reflective of our hypothesis, we use contrast weights of +3 in the CEO/Twitter condition, −1 in the CEO/website condition, −1 in the Investor Relations/website condition, and −1 in

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the Investor Relations/Twitter condition. Results show that the planned contrast is statistically significant ($p = 0.02$, one-tailed), supporting H1.[9]

Follow-up simple effects tests reported in table 1, panel C, reveal that investors are more willing to invest when the CEO discloses firm news via a Twitter account compared to when the same news is disclosed via the Investor Relations Twitter account ($p = 0.04$, one tailed), or from the CEO on a website ($p = 0.04$, one-tailed). Further, investors' willingness to invest judgments are insensitive to Twitter or website disclosure medium when the message source is Investor Relations ($p = 0.99$, two-tailed). Finally, investors' willingness to invest judgments are insensitive to message source when the information is disclosed via a website ($p = 0.87$, two-tailed).

4.3 TEST OF H2

Our second hypothesis predicts that investor trust (our operational measure of a social bond) mediates the moderating effect of disclosure medium on the relation between message source and investment decisions. Given our expectation of mediated moderation, we test H2 using a two-group structural equations modeling (SEM) approach that simultaneously estimates separate coefficients for the CEO and Investor Relations conditions. The model comprises both a measurement model and a structural model, which is graphically depicted in figure 3, panel A.

As shown in figure 3, panel A, the measurement model depicts a second-order confirmatory factor analysis (CFA). The measurement model depicts the links between the set of 10 questions we use to measure trust (presented in table 2, panel A) and the first-order affective and cognitive trust factors.[10] Consistent with theory, we model the five affective (cognitive) trust questions with a nonzero loading on the affective (cognitive) trust factor they were designed to measure and a zero loading on the other trust factor. The measurement model also depicts the links between the two first-order factors and a second-order trust factor. The second-order overall trust factor reflects trust having both affective and cognitive components. Table 2, panel B, reports descriptive statistics for responses to each trust question by condition.

Figure 3, panel A, also depicts the structural model we use to test H2. We present predictions and results for the Investor Relations conditions (in parentheses) immediately below results for the CEO conditions. H2

[9] A semiomnibus test (untabulated) confirms that the residual variance attributable to main and interactive effects of disclosure medium and message source after accounting for our planned contrast is not significant ($F = 0.25$, $p = 0.86$).

[10] In an out-of-sample survey, we asked participants the extent to which they "perceive trust in the CEO is essentially the same as trust in the firm" on a 101-point scale with endpoints 0 ("strongly disagree") and 100 ("strongly agree"). The mean response of 70.39 across all conditions is significantly greater than the scale midpoint of 50 (p-value < 0.01), suggesting participants believe trust in the CEO is essentially the same as trust in the firm. As a result, we ask participants in all conditions about their trust in the CEO as opposed to trust in the firm's Investor Relations.

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TABLE 2
Descriptive Statistics: Trust

Panel A: Questions used to measure trust

Affective Trust
1. If I share my concerns with Dan Athens, I feel he would listen.
2. When making important decisions, I feel Dan Athens acts in the best interest of me and other Armano shareholders.
3. I would feel a sense of personal loss if Dan Athens left Armano.
4. In the future, I feel I could count on Dan Athens to consider how his decisions and actions will affect me and other Armano shareholders.
5. I feel Dan Athens would display a warm and caring attitude towards me and other Armano shareholders.

Cognitive trust
6. Dan Athens approaches his job with professionalism and dedication
7. Given Dan Athens' track record, I see no reason to doubt his competence and preparation for the job.
8. I can rely on Dan Athens not to make my job of evaluating Armano more difficult by careless work.
9. Most people, even those who are not close friends of Dan Athens, likely trust and respect him.
10. Others who must interact with Dan Athens likely consider him to be trustworthy.

Panel B: Measures of trust, mean (standard error), $n = 151$

	Experimental Condition (Medium × Message Source)			
Question	Website/IR ($n = 36$)	Website/CEO ($n = 51$)	Twitter/IR ($n = 30$)	Twitter/CEO ($n = 34$)
Affective trust				
1	50.56 (3.67)	50.00 (2.96)	52.00 (4.11)	60.00 (4.18)
2	55.83 (2.83)	59.02 (2.34)	61.67 (3.59)	65.88 (3.22)
3	49.17 (3.66)	50.98 (3.08)	42.33 (4.23)	57.94 (4.00)
4	53.33 (3.19)	60.39 (2.38)	59.00 (3.94)	63.82 (3.10)
5	57.78 (2.85)	58.63 (2.21)	58.00 (4.11)	66.76 (3.91)
Cognitive trust				
6	68.06 (2.51)	64.71 (2.33)	68.33 (3.69)	72.65 (2.36)
7	60.56 (2.36)	62.75 (2.48)	67.33 (3.68)	69.41 (2.42)
8	54.72 (2.66)	60.78 (2.17)	53.67 (4.46)	65.59 (3.19)
9	58.33 (2.47)	60.98 (2.55)	56.67 (3.47)	68.53 (2.47)
10	60.28 (1.97)	62.35 (2.35)	57.67 (3.61)	71.76 (2.37)

This table reports descriptive statistics for participants' responses to our 10 questions used to measure trust by condition. Panel B of this table presents means and standard errors for the 10 questions by condition. Participants respond to all questions on a 101-point scale ranging from 0 = strongly disagree to 100 = strongly agree. We manipulate disclosure medium by providing a link to firm press releases directly from a website or from Twitter. We manipulate message source by providing the information from Investor Relations (IR) or the CEO.

predicts the following relations for the CEO conditions: (1) a significant positive relation between disclosure medium and trust (Link 1), a significant positive relation between trust and final willingness to invest (Link 3), and a reduction in the significance (partial mediation) or an insignificant relation (full mediation) between disclosure medium and final willingness to invest (Link 2).

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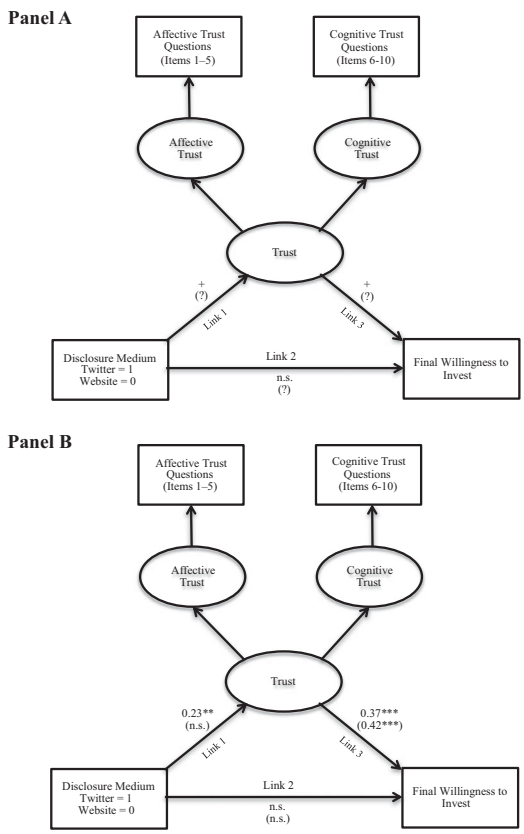


FIG. 3.—Trust as a mediator of the influence of disclosure medium and message source on willingness to invest—test of H2 using two-group structural equations model. This figure presents our predicted (panel A) and observed (panel B) coefficients for H2. For each link, predicted signs and observed coefficients not in parentheses (in parentheses) are for the CEO (firm Investor Relations) source condition. The overall model's comparative fit index (CFI) is 0.95, Bollen–Stine bootstrap $p = 0.16$, SRMR of 0.06, and RMSEA of 0.05. Models with CFI > 0.95, $p > 0.05$, SRMR < 0.09, and RMSEA < 0.05 are considered good fits (Marsh, Hau, and Wen [2004], Iacobucci [2010]). The preliminary willingness to invest variable is not normally distributed, which inflates the chi-square test statistic of overall model fit, causing an inflated Type 1 error rate for rejecting the model as a good fit of the data. We correct for nonnormality by conducting the Bollen–Stice bootstrapping process and using the Bollen–Stine p-value rather than the maximum likelihood p-value to assess the model fit (Bollen and Stine [1993]). Table 1 presents descriptive statistics for preliminary and final willingness to invest. Table 2 presents descriptive statistics for the 10 trust questions. Table 3 presents results for the measurement model. Not shown in the figure but included in our model, we find a significant effect of preliminary willingness to invest on trust ($p < 0.01$) and on final willingness to invest ($p < 0.01$) for both message source groups. *, **, *** indicate one-tailed significance (given our directional predictions) at less than 0.10, 0.05, and 0.01 levels, respectively.

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TABLE 3
Measurement Model Descriptive Statistics

Item	Standardized Estimate	Standard Error	Critical Ratio	*p*-Value
Item 1 ← - - Affective trust	0.71	0.16	6.86	<0.001
Item 2 ← - - Affective trust	0.74	0.12	7.18	<0.001
Item 3 ← - - Affective trust	0.57	0.16	5.34	<0.001
Item 4 ← - - Affective trust	0.70	0.12	6.68	<0.001
Item 5 ← - - Affective trust	0.78	*	*	*
Item 6 ← - - Cognitive trust	0.80	*	*	*
Item 7 ← - - Cognitive trust	0.79	0.13	8.00	<0.001
Item 8 ← - - Cognitive trust	0.72	0.14	7.05	<0.001
Item 9 ← - - Cognitive trust	0.79	0.13	7.60	<0.001
Item 10 ← - - Cognitive trust	0.78	0.13	7.85	<0.001
Affective Trust ← - - trust	0.99	1.78	8.25	<0.001
Cognitive Trust ← - - trust	0.99	1.47	8.53	<0.001

This table reports results of the measurement model portion of the structural equations model reported in figure 2, panel A. Table 2, panel A, reports the 10 questions. We constrain one affective trust (item 5) and one cognitive trust (item 6) factor-loading parameter to 1.0 (i.e., a reference variable) to ensure the latent variable scales are determined (Byrne [2010]). AMOS graphics automatically assigns this value to an indicator variable, but the model is robust to reassigning any other indicator variables as the reference variable. We manipulate disclosure medium by providing a link to firm press releases directly from a website or from Twitter. We manipulate message source by providing the information from Investor Relations (IR) or the CEO.

Table 3 reports the results of the measurement model. As expected, all factor loadings are significant (all p-values < 0.001) and modification indices do not indicate any cross-factor loadings (untabulated). Figure 3, panel B, presents the results of the structural model, which are consistent with our expectations. Specifically, for the CEO conditions, the relation between disclosure medium and trust (Link 1) is positive and significant ($p = 0.02$, one-tailed). The relation between trust and final willingness to invest (Link 3) is positive and significant ($p < 0.01$). Finally, the (previously significant) relation between disclosure medium and final willingness to invest (Link 2) documented in our test of H1, is now insignificant ($p = 0.29$, two-tailed). A Sobel test (untabulated) confirms that, consistent with H2, trust mediates the effect of disclosure medium on willingness to invest ($p = 0.05$, one-tailed) in our CEO conditions.[11,12]

4.4 FOLLOW-UP EXPERIMENT: REPETITION OF NEGATIVE NEWS

One of the unique and naturally occurring features of Twitter is the repetition of information. We conduct a follow-up experiment incorporating this element to (1) increase the generalizability of our findings by

[11] A nested model test reveals that our model is a better fit than a model that constrains all coefficients between the two source groups to be equal ($\chi^2 = 55.17$, $p < 0.01$), which is evidence of an interaction.

[12] Results for all hypotheses are robust to including participants' assessments of firm risk and management's forthcomingness, gender, first language, information acquisition, general propensity to trust, general Twitter use, experience with financial statements, and investing as controls.

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providing evidence that the development of a social bond between investors and a CEO on Twitter is robust to repetitive negative news and (2) examine if the repetition of information common to Twitter is a potential explanation for the higher market sensitivity to negative earnings surprises for firms active on Twitter. Participants are 194 senior accounting students. In the follow-up experiment, after participants in the Twitter conditions view each press release, they are automatically pushed an additional three to seven tweets (depending on the press release), directly quoting phrases from the press release. Participants in the website conditions view the same press releases, but do not receive the repeated information.

Prior research shows that stronger social bonds are resilient to even repeated instances of negative news, as individuals often respond in associative ways that maintain or strengthen the social bond (Khalifa and Shukla [2014]). As a result, we expect the relative strength of the social bond to determine how investors react to repeated news in our experiment. In the presence of a relatively weak social bond, as in our Investor Relations/Twitter condition, we expect that repetitive negative news will amplify assessments of the severity of that news, damage trust to a greater degree (Chen, Saparito, and Belkin [2011]), and result in a more extreme negative reaction (Kasznik and Lev [1995], Libby and Tan [1999], Joe [2003]). In the presence of a relatively strong social bond and more enduring trust, as in our CEO/Twitter condition, we expect repetitive negative news to have little incremental effect on investor judgments.

Figure 2, panel B, displays the pattern of cell means for our dependent variable. We test our expectation using contrast weights of +4 in the CEO/Twitter condition, −1 in the CEO/website condition, −1 in the Investor Relations/website condition, and −2 in the Investor Relations/Twitter condition.[13] Table 4 reports results from our follow-up experiment, which are consistent with the results for H1 reported for our primary experiment. Specifically, as reported in panel C, the planned contrast is statistically significant ($p < 0.01$), and follow-up simple effects tests reveal that investors are more willing to invest when the CEO discloses firm news via a Twitter account than via the Investor Relations Twitter account ($p < 0.01$, one tailed), or the CEO on a website ($p = 0.06$, one-tailed).[14] Further, investors' willingness to invest is insensitive to message source when news is disclosed via a website ($p = 0.83$, two-tailed). Finally, in contrast to our primary experiment (but consistent with expectations given

[13] We use a different set of contrast codes in the follow-up study from those in our primary experiment because when we add the repeated feature of Twitter we expect the Investor Relations Twitter condition to be the lowest cell in terms of investment. This expectation is driven by the repeated negative news in the presence of a weaker social bond.

[14] A semiomnibus test (untabulated) confirms that the residual variance attributable to main and interactive effects of disclosure medium and message source after accounting for our planned contrast is not significant ($F = 0.37$, $p = 0.77$). Results are robust to alternative contrast code specifications maintaining our predicted ordering of cells (i.e., +2, +1, +1, −4).

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TABLE 4

Follow-Up Experiment: How Disclosure Medium and Message Source Affect Willingness to Invest: Tests of H1

Panel A: Unadjusted and adjusted willingness to invest, mean (standard error), $n = 194$

	Experimental Condition			
	Website/IR ($n = 56$)	Website/CEO ($n = 50$)	Twitter/IR ($n = 42$)	Twitter/CEO ($n = 46$)
Unadjusted preliminary willingness to invest	68.84 (2.34)	70.60 (2.53)	67.74 (2.50)	72.83 (2.17)
Unadjusted final willingness to invest	50.36 (2.38)	50.40 (2.24)	41.67 (2.58)	56.52 (2.79)
Adjusted final willingness to invest	50.82 (2.15)	50.16 (2.27)	42.57 (2.49)	55.39 (2.38)

Panel B: ANCOVA model of final willingness to invest

Source of Variation	SS	df	MS	*F*-Statistic	*p*-Value
Disclosure medium	109.27	1	109.27	0.42	0.52
Message source	1,752.84	1	1,752.84	6.78	0.01
Disclosure medium × Message source	2,173.08	1	2,173.08	8.41	<0.01
Preliminary willingness to invest	8,529.51	1	8,529.51	33.01	<0.01
Error	48,832.20	189	258.37		

Panel C: Planned contrast coding and follow-up simple effects tests for final willingness to invest

Source of Variation	df	*F*-Statistic	*p*-Value
Overall tests:			
Twitter/CEO greater than Website/CEO, Website/IR, or Twitter/IR. Contrast weights (4, –1, –1, –2)	1	28.64	<0.01†
Follow-up simple effects tests:			
Effect of message source given Twitter disclosure medium	1	13.81	<0.01†
Effect of message source given Website disclosure medium	1	0.05	0.83
Effect of disclosure medium given firm Investor Relations message source	1	6.32	<0.01†
Effect of disclosure medium given CEO message source	1	2.53	0.06†

This table presents descriptive statistics, an ANCOVA model, and simple main effects tests for participants' willingness to invest. Participants rated (1) the attractiveness of an investment in the firm's stock and (2) the likelihood that they would invest in the firm's stock. The willingness to invest variable is simply an average of these measures. Our dependent variable, adjusted final willingness to invest, is final willingness to invest adjusted for preliminary willingness to invest as a covariate. The simple effects and contrast *F*-statistic are calculated using the ANCOVA error term and adjusted means. In our experiment, we manipulate disclosure medium by providing a link to firm press releases directly from a website or from Twitter. We manipulate message source by providing the information from Investor Relations (IR) or the CEO. †*p*-values are one-tailed equivalents for directional predictions.

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we manipulate the repetition of negative news), investors are less willing to invest when firm news is disclosed via the Investor Relations Twitter account than via the Investor Relations website ($p < 0.01$, one-tailed). This suggests that, in the presence of a relatively weak social bond, investors are negatively impacted by repetitive negative news.[15–17]

We also provide (untabulated) qualitative evidence of our hypotheses tests by examining the key factors participants list in support of their investment decision. Consistent with the formation of a stronger social bond, participants receiving news from the CEO via Twitter are significantly more likely to mention the CEO's reputation or the frequency of communications (24/46 or 52%) than participants in any of the other conditions (10%–30% of participants in each condition; χ^2 test, all p-values < 0.02). They are also more likely to discount subsequent negative news as a one-time event (21/46 or 46%) than are participants in any of the other conditions (8%–12%; χ^2 test, all p-values < 0.01). Conversely, participants in the Investor Relations/Twitter condition are more likely to refer to the poor performance as persistent in nature (18/42 or 43%) than are participants in any of the other conditions (12%–24%; χ^2 test, all p-values < 0.03).

[15] We confirm that any differences in information acquisition caused by the repeated information in Twitter conditions are *not* driving our results. We measure information acquisition using five true/false questions that ask about the five pieces of information tweeted from the last press release containing the negative earnings surprise. Though participants in Twitter conditions answer more recall questions correctly than participants in website conditions (average correct of $4.3 > 4.0$; $p = 0.02$, two-tailed), including information acquisition scores as a control variable in our tests of H1 and H2 does not change the statistical significance of our results.

[16] A possible alternative explanation is that investors are more willing to invest in the CEO/Twitter condition because they are buying anticipating a negative price overreaction after the bad news. To rule out this alternative explanation, we first examine the key factors participants list supporting their investment decision. Only six participants mention justifications like "the stock value probably decreased . . . Buy low and sell high," which does not differ across conditions (all χ^2 test p-values > 0.24, two-tailed). Second, we examine the change in willingness to invest after the negative shock. In all four conditions, participants on average decrease their final from their preliminary willingness to invest after the negative shock (average by condition ranges from –15.91 to –23.89). Only 19 participants *increase* their willingness to invest judgments after the negative shock, which does not differ across conditions (all χ^2 test p-values > 0.72 two-tailed). Excluding the six participants who mention a value investing strategy or the 19 participants who increase their willingness to invest after the negative shock does not change the statistical significance of any of our results.

[17] We also rule out a possible alternative explanation that the message source affects perceptions of the credibility of information and thus investors' reaction to repeated information. Specifically, a potential alternative explanation is that an official Investor Relations source is perceived as more credible than the personal CEO source, thus causing investors to be least willing to invest when firm news is repeated via the Investor Relations Twitter account. We rule out this alternative explanation by finding no differences in the perceived credibility of information across CEO and Investor Relations message sources (all p-values > 0.51).

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4.5 FURTHER SUPPORT FOR OUR THEORY: EVIDENCE OF ENDURING TRUST AND A REPLICATION

In this section, we provide further support for our theory by reporting two additional experiments: a supplemental experiment varying the valence of news and a replication experiment that confirms our initial results. Our supplemental experiment provides evidence that social bonds facilitate the development of more *enduring* trust, and provides evidence supporting our choice to focus on a negative news setting in our primary experiment. Participants in our supplemental experiment are 167 senior accounting students. The experiment uses a 2 × 3 between-subjects design with disclosure medium (website vs. Twitter with *no* repeated Tweets) and earnings surprise (negative vs. neutral vs. positive) as our manipulated independent variables. We hold constant that the message source is the CEO. The experimental materials and procedure are identical to our primary experiment, with the exception that the final press release reflects negative, neutral, or positive news.

Table 5 reports results. Consistent with our theory, simple effects tests reported in panel C show that investors' willingness to invest judgments are insensitive to Twitter or website disclosure following a positive or a neutral earnings surprise (both p-values > 0.41, two-tailed). However, investors discount the negative surprise and are more willing to invest when the CEO discloses the news via a Twitter account relative to a website ($p = 0.02$, one-tailed). In fact, investors receiving news from the CEO on Twitter discount the negative surprise to such an extent that their reaction does not differ from their reaction to neutral news ($p = 0.66$, two-tailed).

Also consistent with our theory, in untabulated analyses, we find that investors display similar levels of overall trust (computed as the average of our 10 trust questions) when the disclosure medium is either Twitter or a website following a positive ($65.89 \approx 62.57$, $p = 0.31$, two-tailed) and a neutral surprise ($64.99 \approx 64.14$, $p = 0.81$, two-tailed). Perhaps most important, only in the condition where the CEO has built more enduring trust by communicating via Twitter is there no difference in trust between a negative, neutral, or positive surprise ($63.33 \approx 64.99 \approx 65.89$, all p-values > 0.40, two-tailed). In all other conditions, trust is lower following a negative surprise than following a neutral ($56.20 < 64.14$, $p = 0.01$, one-tailed) or positive surprise ($56.20 < 62.57$, $p = 0.04$, one-tailed). Taken together, results from our supplemental experiment provide evidence that social bonds facilitate individuals developing more *enduring* trust, and that the cleanest setting to test and observe such differences is when a company reports negative news.

In our second additional experiment, we increase the external validity of our findings by replicating them using an alternative participant group of 185 Amazon MTurk workers (see Trochim, Donnelly, and Arora [2016] and Open Science Collaboration [2015] in *Science* for a discussion of the

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TABLE 5

Supplemental Experiment: How Disclosure Medium and Earnings Surprise Affect Willingness to Invest

Panel A: Unadjusted and adjusted willingness to invest, mean (standard error), $n = 167$

	Experimental Condition					
	Website/ Negative Surprise ($n = 21$)	Website/ Neutral Surprise ($n = 28$)	Website/ Positive Surprise ($n = 28$)	Twitter/ Negative Surprise ($n = 37$)	Twitter/ Neutral Surprise ($n = 22$)	Twitter/ Positive Surprise ($n = 31$)
Unadjusted preliminary willingness to invest	73.91 (2.13)	76.52 (2.50)	73.96 (1.97)	73.28 (2.58)	75.48 (2.34)	70.79 (2.86)
Unadjusted final willingness to invest	53.67 (4.10)	63.70 (2.89)	75.80 (2.80)	61.50 (2.95)	61.07 (4.03)	70.63 (3.00)
Adjusted final willingness to invest	55.76 (3.46)	62.04 (2.86)	75.73 (2.85)	60.39 (2.42)	60.06 (3.22)	72.52 (2.72)

Panel B: ANCOVA model of final willingness to invest

Source of Variation	SS	df	MS	*F*-Statistic	*p*-Value
Disclosure medium	1.42	1	1.42	0.01	0.94
Earnings surprise	8,074.14	2	4,037.07	17.72	<0.01
Disclosure medium × Earnings surprise	470.01	2	235.01	1.03	0.36
Preliminary willingness to invest	10,876.79	1	10,876.79	47.75	<0.01
Error	36,447.57	160	227.80		

Panel C: Simple effects tests for final willingness to invest

Source of Variation	df	*F*-Statistic	*p*-Value
Website/Positive surprise > Website/Neutral surprise	1	11.68	<0.01†
Website/Neutral surprise > Website/Negative surprise	1	3.78	0.03†
Twitter/Positive surprise > Twitter/Neutral surprise	1	8.84	<0.01†
Twitter/Neutral Surprise vs. Twitter/Negative Surprise	1	0.20	0.66
Website/Negative surprise < Twitter/Negative surprise	1	4.04	0.02†
Website/Neutral surprise vs. Twitter/Neutral surprise	1	0.22	0.64
Website/Positive surprise vs. Twitter/Positive surprise	1	0.67	0.41

This table presents descriptive statistics, an ANCOVA model, and simple main effects tests for participants' willingness to invest. Participants rated (1) the attractiveness of an investment in the firm's stock and (2) the likelihood that they would invest in the firm's stock. The willingness to invest variable is simply an average of these measures. Our dependent variable, adjusted final willingness to invest, is final willingness to invest adjusted for preliminary willingness to invest as a covariate. In our experiment, we manipulate disclosure medium by providing a link to firm press releases directly from a website or from Twitter and hold constant across conditions the message source as the CEO. We manipulate earnings surprise by providing a final press release that reflects a positive, neutral, or negative earnings surprise.
† *p*-values are one-tailed equivalents for directional predictions.

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TABLE 6

Replication Experiment: How Disclosure Medium and Message Source Affect Willingness to Invest: Tests of H1

Panel A: Unadjusted and adjusted willingness to invest, mean (standard error), $n = 185$

	Experimental Condition					
	Website/IR ($n = 41$)	Website/CEO ($n = 27$)	No Repeat Tweets Twitter/IR ($n = 35$)	No Repeat Tweets Twitter/CEO ($n = 29$)	Repeat Tweets Twitter/IR ($n = 26$)	Repeat Tweets Twitter/CEO ($n = 27$)
Unadjusted preliminary willingness to invest	79.76 (2.02)	79.63 (3.85)	72.57 (3.44)	79.66 (3.07)	75.58 (2.76)	83.52 (2.23)
Unadjusted final willingness to invest	53.90 (3.31)	53.70 (4.58)	51.71 (3.83)	62.76 (4.02)	46.92 (3.10)	64.26 (3.95)
Adjusted final willingness to invest	52.84 (2.73)	52.74 (3.37)	55.98 (2.99)	61.77 (3.25)	48.96 (3.44)	60.41 (3.39)

Panel B: ANCOVA model of final willingness to invest: No repeat tweets

Source of Variation	SS	df	MS	*F*-Statistic	*p*-Value
Disclosure medium	1,214.66	1	1,214.66	3.82	0.05
Message source	241.12	1	241.12	0.76	0.39
Disclosure medium × Message source	259.06	1	259.60	0.82	0.18†
Preliminary willingness to invest	22,890.27	1	22,890.27	72.06	<0.01
Error	40,341.42	127	317.65		

Panel C: Planned contrast coding and follow-up simple effects tests for final willingness to invest: No repeat tweets

Source of Variation	df	*F*-Statistic	*p*-Value
Overall tests:			
Twitter/CEO greater than Website/CEO, Website/IR, or Twitter/IR. Contrast weights (3, –1, –1, –1)	1	4.60	0.02†
Follow-up simple effects tests:			
Effect of message source given Twitter disclosure medium	1	1.71	0.10†
Effect of message source given Website disclosure medium	1	0.001	0.98
Effect of disclosure medium given firm Investor Relations message source	1	0.60	0.44
Effect of disclosure medium given CEO message source	1	3.74	0.03†

Panel D: ANCOVA model of final willingness to invest: Repeat tweets

Source of variation	SS	df	MS	*F*-Statistic	*p*-Value
Disclosure medium	103.88	1	103.88	0.32	0.58
Message source	1,012.85	1	1,012.85	3.07	0.08
Disclosure medium × Message source	1,050.80	1	1,050.80	3.19	0.04†
Preliminary willingness to invest	11,568.46	1	11,568.46	35.06	<0.01
Error	38,275.81	116	329.96		

(*Continued*)

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TABLE 6—*Continued*

Panel E: Planned contrast coding and follow-up simple effects tests for final willingness to invest: Repeat tweets

Source of Variation	df	*F*-Statistic	*p*-Value
Overall Tests:			
Twitter/CEO greater than Website/CEO, Website/IR, or Twitter/IR. Contrast weights (4, –1, –1, –2)	1	3.91	0.03†
Follow-up simple effects tests:			
Effect of message source given Twitter disclosure medium	1	5.58	<0.01†
Effect of message source given Website disclosure medium	1	0.001	0.98
Effect of disclosure medium given firm Investor Relations message source	1	0.78	0.19†
Effect of disclosure medium given CEO message source	1	2.59	0.05†

Panel F: Additional planned comparison tests for final willingness to invest: No repeat tweets compared to repeat tweets

Source of Variation	df	*F*-Statistic	*p*-Value
Effect of repeated tweets given CEO message source and Twitter disclosure medium	1	0.08	0.77
Effect of repeated tweets given firm Investor Relations message source and Twitter disclosure medium	1	2.78	0.05†

This table presents descriptive statistics, ANCOVA models, and simple main effects tests for participants' willingness to invest for our Amazon MTurk replication experiment. Participants rated (1) the attractiveness of an investment in the firm's stock and (2) the likelihood that they would invest in the firm's stock. The willingness to invest variable is simply an average of these measures. Our dependent variable, adjusted final willingness to invest, is final willingness to invest adjusted for preliminary willingness to invest as a covariate. The simple effects and contrast *F*-statistic are calculated using the ANCOVA error term and adjusted means. In our experiment, we manipulate disclosure medium by providing a link to firm press releases directly from a website or from Twitter. We manipulate message source by providing the information from Investor Relations (IR) or the CEO.
† *p*-values are one-tailed equivalents for directional predictions.

benefits of reproducibility).[18] Our replication experiment uses a 2 × 2 + 2 design, with disclosure medium (website vs. Twitter with *no* repeated Tweets) and message source (firm Investor Relations vs. CEO) as our manipulated independent variables. We also include two additional Twitter disclosure medium conditions in which participants receive repeated tweets. The experimental materials and procedure are identical to our primary experiments with the final press release containing negative news.

[18] Downs et al. [2010] stress the need for screening questions to disqualify MTurk respondents who quickly complete tasks for compensation, compromising data quality. Our initial sample included 240 Amazon Turk workers. We remove 25 participants with the same IP address as other responses, 5 participants who finished the experiment in less than five minutes, and 25 participants who ignored our Amazon Turk recruiting instructions asking for participants with investing experience.

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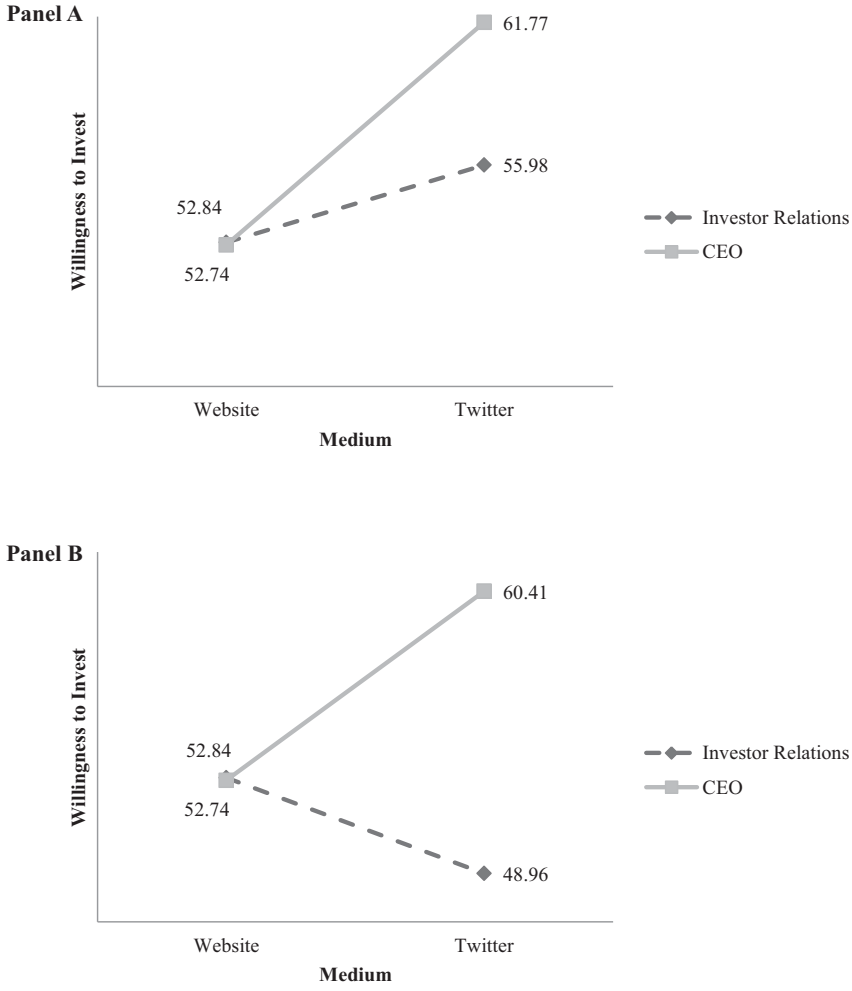


FIG. 4.—The effect of disclosure medium and message source on investors' willingness to invest. This figure graphically depicts our observed mean values for participants' final willingness to invest (adjusted for preliminary willingness to invest as a covariate). Panel A (panel B) depicts means for the replication of our primary experiment (the follow-up experiment) in which our Twitter manipulation does not (does) contain repeated tweets. Table 6 presents descriptive statistics for preliminary and final willingness to invest for the replications of both the primary experiment and the follow-up experiment.

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Table 6 reports results, and figure 4 displays the pattern of cell means for our dependent variable.

Applying the same contrast codes used in our primary experiment, we replicate support for H1. Specifically, as seen in table 6, panel C, the planned contrast is statistically significant ($F = 4.60$, $p = 0.02$, one-tailed), and the simple effects tests reveal that investors are more willing to invest when the CEO discloses news via a Twitter account than when Investor Relations discloses news via a Twitter account ($p = 0.10$, one tailed), or when the CEO discloses news via a website ($p = 0.03$, one-tailed). We also replicate support for H2, which predicts that investor trust mediates the moderating effect of disclosure medium on the relation between message source and investment decisions. Consistent with our primary results, our replicated SEM analysis reveals the relation between disclosure medium and trust is positive and significant ($p = 0.01$, one-tailed), the relation between trust and final willingness to invest is positive and significant ($p < 0.01$, one-tailed), and the (previously significant) relation predicted in H1 between disclosure medium and final willingness to invest is now insignificant ($p = 0.33$, two-tailed).

Finally, we replicate the results of our follow-up study. Applying the same contrast codes used in our follow-up experiment, table 6, panel E, shows that the planned contrast is statistically significant ($F = 3.91$, $p = 0.03$, one-tailed). Table 6, panel F, shows that, once again, investors are unaffected by whether negative news is presented once or repeated when the source is the CEO's Twitter account ($p = 0.77$, two-tailed), but are significantly negatively affected by the repetition when the source is the Investor Relations Twitter account ($p = 0.05$, one-tailed). These results further support that the development of a strong social bond with the CEO via Twitter is robust to repeated negative news, but that, without this social bond, the repetition of negative news leads to a negative reaction.

5. *Conclusion*

In this study, we provide experimental evidence that when a CEO uses Twitter to communicate with investors, it facilitates the creation of a social bond and a more enduring form of trust, which subsequently influences how investors perceive the CEO and the news the CEO reports. Specifically, in our primary experiment, we show that investors trust the CEO more and are more willing to invest in the firm when the CEO communicates firm news followed by a negative earnings surprise through a personal Twitter account than when the news and surprise comes from the CEO via a website or from the firm's Investor Relations Twitter account or website. In our follow-up experiment, we further show that repeating the negative news does not additionally affect investors who received the news from the CEO's Twitter account, but does further negatively impact investors who received the news via other disclosure mediums, especially those who received the news via the Investor Relations Twitter account.

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Our study has several limitations that could be explored in future research. First, our study focuses on one type of social media: Twitter. We do not examine other types of social media. Second, we did not examine how managers might strategically time their Twitter activity to influence investors' judgments. Third, we only examine tweets that originate from the company or from the CEO. We do not examine the vast number of other users who tweet corporate news. Fourth, we randomly assign participants to receive tweets from either the CEO or the company's Investor Relations. In the real world, investors can choose to receive news from one account or both accounts. Chen, Hwang, and Liu [2017] find that CEOs or CFOs who communicate via a personal Twitter account grow their retail shareholder base by 22%, even when controlling for company Twitter account activity. This finding suggests that the benefits of CEOs communicating via a personal Twitter account are likely incremental to the benefits of a company communicating via a corporate Twitter account. Fifth, we focus on an important negative event (missing an earnings target), but not as significant of a negative event (e.g., fraud or a restatement). Sixth, we chose one level of repeated negative information for Twitter conditions in our follow-up experiment. An extreme level of repeated negative news likely exists at which our results no longer hold. Finally, we focus on a benefit and do not examine potential costs of CEOs communicating with investors via Twitter.

Our study makes several contributions. By using an experiment, we are able to hold the amount, content, and valence of information constant and provide empirical evidence of a potential benefit of CEOs disclosing information via Twitter that cannot be examined using archival data. Specifically, we hold constant factors that prior research identifies as important for investor reactions to earnings announcements, including the content and volume of other users' tweets (Lee, Hutton, and Shu [2015]) and managers' strategic tweeting decisions (Bhagwat and Burch [2016]). Finally, using an experiment, we can measure the underlying process (i.e., the formation of a social bond that leads to enduring trust) that influences investors' judgments. Overall, our study contributes to recent literature investigating the consequences of social media (e.g., Sprenger et al. [2013], Blankespoor, Miller, and White [2014], Chen et al. [2014], Chen, Hwang, and Liu [2017]) and the disclosure medium literature (e.g., Elliott, Hodge, and Sedor [2012]), and has practical implications for managers. Many CEOs express reluctance to communicate with stakeholders via Twitter due to the perceived time commitment and potential legal ramifications of unscripted tweets going viral (Kwoh and Korn [2012]). Our results suggest CEOs could reap potential benefits from establishing a Twitter account and simply tweeting already scripted information from press releases.

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APPENDIX

Manipulations

Panel A: Website disclosure provided by Investor Relations



Panel B: Website disclosure provided by CEO



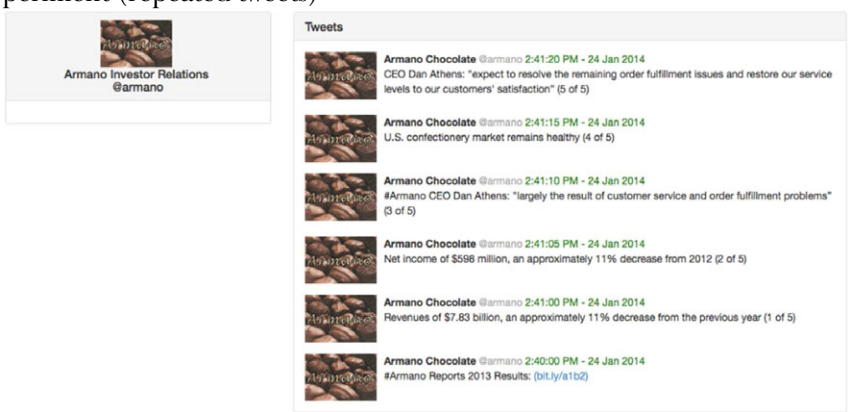
Panel C: Twitter disclosure provided by Investor Relations—primary experiment (no repeated tweets)



Panel D: Twitter disclosure provided by CEO—primary experiment (no repeated tweets)

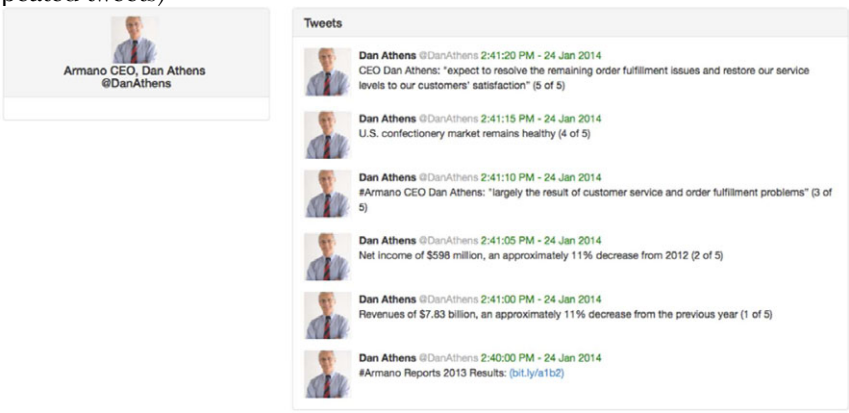


Panel E: Twitter disclosure provided by Investor Relations—follow-up experiment (repeated tweets)



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Panel F: Twitter disclosure provided by CEO—follow-up experiment (repeated tweets)



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